

CRISPR-Cas9 Gene Editing: Mechanisms, Applications, and Challenges

Abstract

CRISPR-Cas9 (Clustered Regularly Interspaced Short Palindromic Repeats — CRISPR-associated protein 9) has emerged as a transformative tool in molecular biology, enabling precise, programmable modifications to genomic DNA across virtually any organism. This report reviews the molecular mechanism of CRISPR-Cas9, its principal applications in research and medicine, and the key technical and ethical challenges that remain.

1. Introduction

The ability to edit genomes with precision has long been a central goal of molecular biology. Earlier technologies — zinc-finger nucleases (ZFNs) and transcription activator-like effector nucleases (TALENs) — established proof-of-concept but were costly and technically demanding to engineer [1]. The adaptation of the bacterial CRISPR-Cas9 system for eukaryotic genome editing, reported in 2012–2013 by Jinek et al. and Cong et al., marked a paradigm shift: a two-component RNA-guided nuclease that can be retargeted to any genomic locus in days [2, 3]. The 2020 Nobel Prize in Chemistry, awarded to Jennifer Doudna and Emmanuelle Charpentier, recognised the foundational importance of this work.

2. Molecular Mechanism

2.1 Native CRISPR Immunity

In bacteria and archaea, CRISPR loci serve as an adaptive immune memory. Foreign DNA sequences are integrated as short "spacers" between palindromic repeats. Upon re-infection, these spacers are transcribed into CRISPR RNA (crRNA), which — complexed with a trans-activating crRNA (tracrRNA) — guides Cas9 to cleave matching viral sequences [4].

2.2 Guide RNA and Target Recognition

For experimental use, the crRNA:tracrRNA duplex is fused into a single guide RNA (sgRNA) of ~100 nucleotides. The 20-nucleotide spacer region at the 5' end directs Cas9 to a complementary DNA sequence (the **protospacer**) immediately upstream of a protospacer adjacent motif (PAM) — 5'-NGG-3' for the most widely used *Streptococcus pyogenes* Cas9 (SpCas9) [2].

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5'-[20 nt protospacer]-NGG-3'    (genomic DNA target strand)
|||||
[20 nt spacer in sgRNA]
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Figure 1. Schematic of sgRNA:DNA base-pairing. Cas9 requires complementarity across the spacer region and recognition of the PAM (NGG) to initiate cleavage.

2.3 DNA Cleavage and Repair

Once bound, the RuvC and HNH nuclease domains of Cas9 each cut one DNA strand, generating a blunt-ended double-strand break (DSB) ~3 bp upstream of the PAM. The cell resolves the DSB via one of two pathways:

- **Non-Homologous End Joining (NHEJ):** Error-prone ligation that frequently introduces small insertions or deletions (indels), causing frameshift mutations useful for gene knockout.
- **Homology-Directed Repair (HDR):** When a donor template is supplied, precise sequence substitution or insertion can be achieved, enabling knock-in edits.

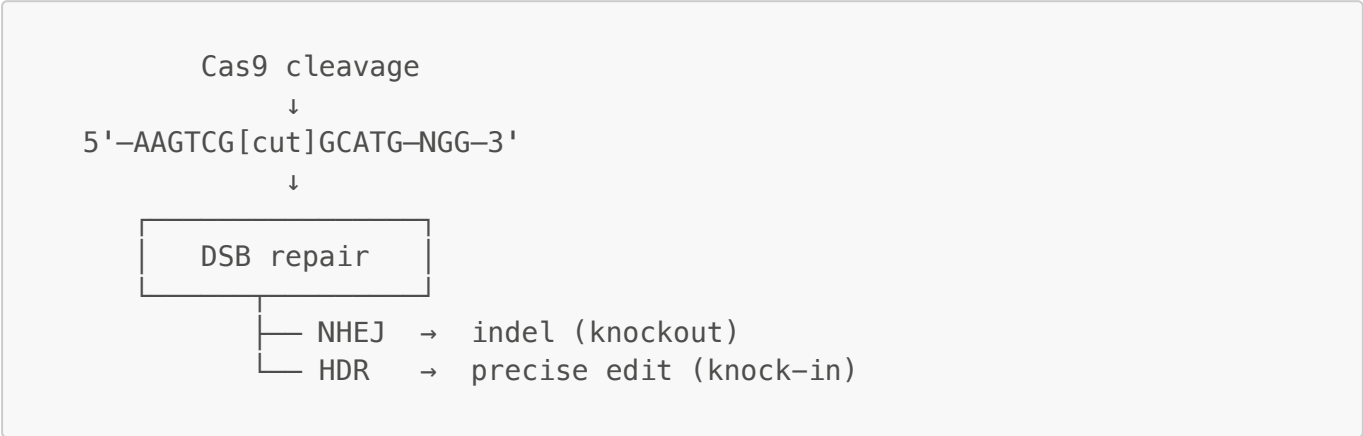


Figure 2. Repair pathway choice after Cas9-induced DSB. NHEJ dominates in most cell types; HDR requires an exogenous template and is more active in dividing cells.

3. Applications

3.1 Functional Genomics

Genome-scale CRISPR knockout screens use pooled sgRNA libraries to systematically disrupt every gene and identify those essential for a phenotype of interest (e.g., drug resistance, cell viability) [5]. This approach has replaced RNA interference as the gold-standard loss-of-function screen due to its higher specificity and complete gene disruption.

3.2 Therapeutic Gene Editing

CRISPR-Cas9 has advanced rapidly into the clinic:

Indication	Strategy	Status (as of 2025)
Sickle cell disease / β -thalassaemia	Reactivate fetal haemoglobin (BCL11A disruption)	FDA-approved (Casgevy, 2023) [6]
Transthyretin amyloidosis	In vivo liver editing to silence TTR	Phase III
Duchenne muscular dystrophy	Exon skipping to restore dystrophin reading frame	Phase I/II
Leukaemia (CAR-T)	Ex vivo editing of T cells to improve persistence	Multiple Phase I

Table 1. Selected clinical-stage CRISPR-Cas9 programmes.

3.3 Base Editing and Prime Editing

Catalytically impaired "nickase" Cas9 variants, fused to deaminase enzymes, enable **base editing** — direct chemical conversion of one nucleotide to another (e.g., C→T or A→G) without a DSB [7]. **Prime editing**, which uses a reverse transcriptase and a prime editing guide RNA (pegRNA), extends this to all 12 types of point mutations and small indels with minimal off-target activity [8].

4. Challenges

4.1 Off-Target Editing

SpCas9 can cleave genomic loci that partially match the sgRNA, raising safety concerns for therapeutic use. Mitigation strategies include high-fidelity Cas9 variants (eSpCas9, HiFi Cas9), truncated guide RNAs, and paired nickases that require two adjacent cuts for a productive DSB [9].

4.2 Delivery

Efficient delivery of Cas9 and sgRNA to target tissues remains a bottleneck. Current strategies include:

- **Ex vivo editing** of harvested cells (most clinically advanced)
- **Adeno-associated virus (AAV)** vectors (limited cargo capacity ~4.7 kb)
- **Lipid nanoparticles (LNPs)** for transient mRNA/RNP delivery (reduced off-target risk)

4.3 Immunogenicity

Pre-existing humoral and cellular immunity to SpCas9 (derived from a common human pathogen) has been detected in a significant fraction of human donors, potentially limiting in vivo applications [10]. Orthologous Cas9 proteins from less common bacteria (e.g., *Staphylococcus aureus*, *Campylobacter jejuni*) offer smaller size and reduced immunogenicity.

4.4 Ethical and Regulatory Considerations

The 2018 report of heritable human embryo editing by He Jiankui drew widespread condemnation and underscored the need for international governance frameworks. Somatic gene therapy is broadly accepted; germline editing remains prohibited in most jurisdictions pending consensus on safety and equity [11].

5. Conclusion

CRISPR-Cas9 has compressed the timeline from genomic hypothesis to experimental validation from months to days, and from bench discovery to clinical therapy faster than any previous editing platform. Key milestones — the first approved CRISPR medicine in 2023 and expanding base/prime editing toolkits — illustrate both the maturity and the ongoing innovation of the field. Realising the full therapeutic potential will require continued progress in delivery technologies, off-target detection methods, and international governance.

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